Scope of the Formal Canonical-Form Reductions Relative to PGVWC07

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1 The source theorem

The translation-invariant canonical-form theorem of Pérez-García, Verstraete, Wolf, and Cirac starts from an arbitrary translation-invariant MPS representation on a finite ring [PGVWC07, Theorem `Th:TIcanonical`, lines 742–763]. It states that the matrices can always be decomposed as

$$A_i = \begin{pmatrix} \lambda_1 A_i^1 & 0 & 0 \\ 0 & \lambda_2 A_i^2 & 0 \\ 0 & 0 & \cdots \end{pmatrix},$$

where $1 \geq \lambda_j > 0$ and each block satisfies:

$$\sum_i A_i^j A_i^{j\dagger} = \mathbf{1}, \qquad \sum_i A_i^{j\dagger} \Lambda^j A_i^j = \Lambda^j,$$

for a diagonal positive full-rank matrix Λ^j, and $\mathbf{1}$ is the only fixed point of

$$\mathcal{E}_j(X) = \sum_i A_i^j X A_i^{j\dagger}.$$

If the initial representation has bond dimension D, the canonical representation has bond dimension at most D.

The statement does not assume injectivity, a condition C1, blockwise primitivity, a prior block decomposition, strict separation of the weights, or that only one block occurs in each weight class. These properties enter elsewhere in the paper as later hypotheses, consequences, or genericity conditions. For example, the injectivity statement appears only after the canonical form has been obtained [PGVWC07, Proposition `prop-inj`, lines 911–950], and the generic uniqueness discussion is a separate theorem [PGVWC07, Theorem `thm-uniq`, lines 1002–1015].

2 Normalization orientation

There is also a convention difference in the direction of the canonical normalization. In the source theorem, the block transfer map is written as

$$\mathcal{E}_j(X) = \sum_i A_i^j X A_i^{j\dagger}$$

and imposes the unital condition

$$\sum_i A_i^j A_i^{j\dagger} = \mathbf{1}.$$

The local canonical-form chapter usually works in the dual trace-preserving, or left-canonical, orientation

$$\sum_i A_i^{j\dagger} A_i^j = \mathbf{1}.$$

This is a convention difference when it is accompanied by the corresponding adjoint transfer map or conjugate-transposed Kraus family. It is nevertheless important to state the convention explicitly: a theorem written only in the left-canonical orientation should not be read literally as PGVWC07 Theorem `Th:TIcanonical` unless the dualization or gauge conversion has also been supplied.

3 The formal reductions

The currently checked statements in the canonical-form reduction chapter prove conditional reductions from prior block hypotheses. In particular, the primitive fixed-point reduction assumes that the blocks are injective, trace-preserving in the chosen gauge, strictly ordered by weight modulus, nonzero, positive-dimensional, and primitive. The peripheral-spectrum variant assumes the same injectivity, normalization, strict weight ordering, nonzero weights, and positive block dimensions, together with the statement that each block transfer map has peripheral spectrum $\{1\}$. The normal-canonical-form reduction starts from a weighted block decomposition already equivalent to the original state and assumes irreducibility, trace-preserving normalization, primitivity, pairwise distinct weight moduli, nonzero weights, and positive bond dimensions.

These are correct conditional consequences, but they are not formalizations of the full PGVWC07 translation-invariant canonical-form theorem.

Their hypotheses are stronger and more structured than the source theorem's hypotheses. Treating them as the full source theorem would import extra assumptions into any later argument that cites canonical-form existence.

4 Formal boundary

The currently checked primitive, peripheral-primitive, and normal-canonical-form reductions should therefore be read as conditional canonical-form theorems rather than as formalizations of the full source theorem.[1] They may be used after their hypotheses have been proved from earlier source-faithful reductions, but they should not be cited as giving canonical form directly from an arbitrary translation-invariant MPS representation.

5 What remains

A faithful formalization of PGVWC07 Theorem `Th:TIcanonical` should start with an arbitrary translation-invariant MPS representation and construct the block decomposition, the positive weights, the unital block gauge, the diagonal full-rank fixed points, and the uniqueness of the identity fixed point, while preserving the stated bond-dimension bound. The conditional primitive, peripheral-primitive, and normal-canonical-form reductions can then be used only after the source argument has produced the corresponding hypotheses or after a later theorem has explicitly assumed them.

References

[PGVWC07] David Pérez-García, Frank Verstraete, Michael M. Wolf, and J. Ignacio Cirac. Matrix product state representations. *Quantum Information and Computation*, 7(5–6):401–430, 2007.

[1] The formal declarations are `MPSTensor.isCanonicalForm_of_primitive`, `MPSTensor.isCanonicalForm_of_peripheralPrimitive`, and `MPSTensor.exists_normalCanonicalForm_of_primitive_blockDecomp`.